ORGANIZED BY

Department of Electrical Engineering

Motilal Nehru National Institute of Technology Allahabad

Technically Co-Sponsored by IEEE UP SECTION (INDIA)

SPECIAL SESSION | SS-04

AI and Digital Twins for Next-Generation Power System Monitoring, Control and Optimization with Renewable Energy Penetration

CALL FOR PAPERS | SPECIAL SESSION

9th Students' Conference on Engineering and Systems

SCES -2027

www.mnnit.ac.in/sces2027

March 12th – 14th, 2027

IMPORTANT DATES

PAPER SUBMISSION DEADLINE
November 15, 2026

After Submission

Acceptance Notification	Registration Open
January 10, 2027	February 05, 2027

ABOUT THE SPECIAL SESSION

The proposed area integrates AI, machine learning, and digital twins to enable intelligent, adaptive, and resilient power and energy systems with high renewable energy penetration. It supports predictive monitoring, real-time control, and optimized operation of renewable energy sources, microgrids, EVs, energy storage, and distributed energy resources.

TOPICS OF INTEREST (but not limited to)

- AI for Renewable-Dominated Power System Monitoring
- Digital Twins for Power and Energy Systems
- AI-Based Renewable Energy Forecasting
- AI-Driven Power System Optimization
- Intelligent Control of Renewable-Dominated Grids
- AI-Enabled Energy Management Systems
- Digital Twins + AI for Predictive Maintenance and Resilience
- Real-Time and Hardware-in-the-Loop Implementation
- Trustworthy and Explainable AI for Power Systems

SPECIAL SESSION ORGANIZER(S)



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PUBLICATION & SUBMISSION

For paper submission, kindly use at the following link:
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